TUMAINI

Sexually transmitted infections and contraceptive use among adolescent girls in Tanzania: a multi-centre prospective cohort

Protocol summary and case report forms for the REDCap homework · Version 3.0, 01-Oct-2026

Study name	TUMAINI (Swahili for “hope”). Replaces the PISTIC study used in 2025.
Design	Prospective cohort, 12 months of follow-up, three follow-up schedules (arms).
Sites (multi-centre)	1 Dar es Salaam (Temeke) · 2 Mwanza · 3 Mbeya. Each site enters and sees only its own participants.
Participants	480 girls aged 16 to 20 years (160 per site) who report having had sex at least once.
Main inclusion criteria	Age 16 to 20; consent (or parent or guardian consent plus assent if younger than 18); locator information; willing to have genital examinations; willing to stay 12 months.
Main exclusion criteria	Plans to move away for more than 3 months in a row; a condition that makes participation unsafe; known HIV infection.
Objectives	1. Prevalence of reproductive tract infections and associated factors at Day 0. 2. Incidence of reproductive tract infections and associated factors during follow-up. 3. Prevalence of contraceptive use and associated factors.
Follow-up schedules	Arm 1 Standard: Day 0 and months 3, 6, 9, 12. Arm 2 Intensive: Day 0, monthly to month 6, then months 9 and 12. Arm 3 Reduced: Day 0 and months 6 and 12 (external genital examination only).
Procedures	Questionnaires on sexual behaviour and contraception (self-completed on a tablet), genital examination, vaginal swabs and blood for STI testing, locator information.
Sponsor	University of Antwerp (fictional study for teaching; no real participants).

Abbreviations

BV	Bacterial vaginosis	**CT**	Chlamydia trachomatis
CVL	Cervicovaginal lavage	**DAG**	Data Access Group (one per site in REDCap)
FGE	Follow-up genital examination form	**FSX**	Sexual history and contraception form
GC	Gonorrhoea (Neisseria gonorrhoeae)	**HSV**	Herpes simplex virus
IUD	Intrauterine device	**LOC**	Locator and knowledge check form (designed by you)
MV	Missed visit form	**PCR**	Polymerase chain reaction
SEGE	Screening genital examination form	**SELIG**	Screening and eligibility form
STI	Sexually transmitted infection	**TV**	Trichomonas vaginalis

Schedule of assessments

Which form is completed at which visit, per arm. Unscheduled visits can happen more than once. A missed visit form is completed for every scheduled visit that is missed.

Arm 1 · Standard follow-up

Form	Day 0	M3	M6	M9	M12	Unsched.	Missed
SELIG · Screening and eligibility	✓						
SEGE · Screening genital examination	✓						
FSX · Sexual history and contraception	S	S	S	S	S	S	
FGE · Follow-up genital examination		✓	✓	✓	✓	✓	
MV · Missed visit							R
LOC · Locator and knowledge check	✓						

Arm 2 · Intensive follow-up

Form	Day 0	M1	M2	M3	M4	M5	M6	M9	M12	Unsched.	Missed
SELIG · Screening and eligibility	✓										
SEGE · Screening genital examination	✓										
FSX · Sexual history and contraception	S	S	S	S	S	S	S	S	S	S	
FGE · Follow-up genital examination				✓			✓	✓	✓	✓	
MV · Missed visit											R
LOC · Locator and knowledge check	✓										

Arm 3 · Reduced follow-up

Form	Day 0	M6	M12	Unsched.	Missed
SELIG · Screening and eligibility	✓				
SEGE · Screening genital examination	✓				
FSX · Sexual history and contraception	S	S	S	S	
FGE · Follow-up genital examination		✓		✓	
MV · Missed visit					R
LOC · Locator and knowledge check	✓				

✓ completed by staff · S survey, completed by the participant on a tablet · R repeating form (one per missed visit)

Visit windows: ±7 days in the first 6 months of the Intensive arm, ±14 days for all other follow-up visits.

SELIG Screening and eligibility

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Completed once, on Day 0, by the study nurse. Decides eligibility and the follow-up arm.

Participant ID	Site	Visit	Date
□ - □□□	□ 1 Dar □ 2 Mwanza □ 3 Mbeya	Day 0	□□/□□□/□□□□

	Date of birth	□□ / □□□ / □□□□ dd mmm yyyy
	Age at screening (years)	__ __ years
Consent and assent		
1a	Did the participant (18 years or older) give written informed consent?	□ Yes □ No *→ Only if the participant is 18 or older. If No: not eligible, end of form.*
1b	Date of informed consent *Only if 1a is Yes*	□□ / □□□ / □□□□ dd mmm yyyy
2a	Did a parent or guardian give written informed consent?	□ Yes □ No *→ Only if the participant is younger than 18. If No: not eligible, end of form.*
2b	Date of parent or guardian consent *Only if 2a is Yes*	□□ / □□□ / □□□□ dd mmm yyyy
3a	Did the participant give written assent?	□ Yes □ No *→ Only if a parent or guardian consented. If No: not eligible, end of form.*
3b	Date of assent *Only if 3a is Yes*	□□ / □□□ / □□□□ dd mmm yyyy
Inclusion criteria: all must be Yes		
4	Is the participant 16 to 20 years old?	□ Yes □ No
5	Has the participant had sex at least once (self-report)?	□ Yes □ No
6	Is the participant (or parent or guardian) willing and able to give locator information?	□ Yes □ No
7	Is the participant willing to have genital examinations?	□ Yes □ No
8	Is the participant willing to stay in follow-up for 12 months?	□ Yes □ No
Exclusion criteria: all must be No		
9	Does the participant plan to move away from the study area for more than 3 months in a row?	□ Yes □ No
10	Does the participant have a condition that, in the opinion of the investigator, makes participation unsafe?	□ Yes □ No
11	Is the participant known to be HIV positive?	□ Yes □ No
Eligibility		
12	Based on items 1 to 11, is the participant eligible for enrolment?	□ Eligible □ Not eligible
13	Was the participant enrolled today?	□ Yes □ No *→ Only if eligible. If Yes, go to item 15.*
14	Reason the eligible participant was not enrolled *Only if 13 is No*	□ Declined enrolment □ Investigator decision □ Did not return for enrolment □ Other
14a	Other reason, specify *Only if 14 is Other*	_____
15	Follow-up schedule (arm) assigned *The record must be placed in this arm in REDCap.* *Only if 13 is Yes*	□ Standard (quarterly) □ Intensive (monthly) □ Reduced (months 6 and 12)

Completed by (initials) □□□□	**Date** □□/□□□/□□□□

SEGE Screening genital examination

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Completed once, on Day 0, by the study clinician after enrolment.

Participant ID	Site	Visit	Date
□ - □□□	□ 1 Dar □ 2 Mwanza □ 3 Mbeya	Day 0	□□/□□□/□□□□

1	First day of last menstrual period	□□ / □□□ / □□□□ dd mmm yyyy
2a	Was a genital examination done?	□ Yes □ No *→ If No, give the reason and end the form.*
2b	Reason the examination was not done *Only if 2a is No*	_____ _____

Signs on examination

3 Were any of these signs seen?

Sign	Answer
a. Redness	□ Yes □ No □ Not applicable
b. Abnormal vaginal discharge	□ Yes □ No □ Not applicable
c. Swelling	□ Yes □ No □ Not applicable
d. Ulcers	□ Yes □ No □ Not applicable
e. Other	□ Yes □ No □ Not applicable

Only if 2a is Yes

3d	Location of ulcers (tick all that apply) *Only if 3 is Yes*	□ Mons □ Labia □ Perineal □ Perianal □ Buttock □ Leg, hip or back
3e	Other sign, specify *Only if 3 is Yes*	_____

Samples collected

4	HSV PCR swab collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
5	Vaginal swab for BV Gram stain collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
6	Vaginal swab for BV storage collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
7	Vaginal swab for GC, CT and TV collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
8	Cervicovaginal lavage collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
9	Archive sample for HSV testing collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable
10	Host DNA sample collected? *Only if 2a is Yes*	□ Yes □ No □ Not applicable

Laboratory results (aetiological diagnosis)

11 Laboratory tests done for the aetiological diagnosis of STIs

Test	Done?	Result
a. Gonorrhoea	□ Yes □ No	□ Negative □ Positive
b. Chlamydia	□ Yes □ No	□ Negative □ Positive
c. Trichomoniasis	□ Yes □ No	□ Negative □ Positive
d. Bacterial vaginosis	□ Yes □ No	□ Negative □ Positive

Completed by (initials) □□□□	**Date** □□/□□□/□□□□

FSX Sexual history and contraception

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Self-completed by the participant on a tablet at every visit (survey). Interviewer reads it aloud if needed.

Participant ID	Site	Visit	Date
□ - □□□	□ 1 Dar □ 2 Mwanza □ 3 Mbeya	_____	□□/□□□/□□□□

This is your visit questionnaire. The questions are about sex and contraception. Sex means vaginal or anal sex, not oral sex. A condom can be a male or a female condom. Your answers are confidential: please answer as honestly as you can.

Sexual history		
1a	Have you ever had sex?	□ Yes □ No → *If No, end of form.*
1b	How old were you the first time you had sex?	□□ years → *Day 0 only.*
1c	Was a condom used the first time? *Only if 1a is Yes*	□ Yes □ No
1d	Was the first time your choice, with your agreement? *Only if 1a is Yes*	□ Yes □ No
1e	Were you offered a reward or money for sex that first time? *Only if 1a is Yes*	□ Yes □ No
1f	How old was the person you first had sex with? *Only if 1a is Yes*	□□ years
1g	How long had you known this person before you had sex? *Only if 1a is Yes*	□□ □ Days □ Months □ Years
The past 3 months		
2a	In the past 3 months, have you had sex?	□ Yes □ No → *If No, go to item 3.*
2b	How many times did you have sex in the past 3 months? *Only if 2a is Yes*	□□□
2c	Of these times, how many times was a condom used? *Only if 2a is Yes*	□□□
2d	How many sexual partners have you had in the past 3 months? *Only if 2a is Yes*	□□
2e	Of these partners, how many were new? *Only if 2a is Yes*	□□
Contraception		
3a	In the past 3 months, have you used any method to prevent pregnancy?	□ Yes □ No → *If No, end of form.*
3b	Which methods did you use? Tick all that apply. *Only if 3a is Yes*	□ Pill □ Injectable □ Implant □ IUD □ Condoms □ Emergency pill □ Safe days □ Other
3c	Other method, specify *Only if 3b = Other ticked*	_____
3d	For how many months have you used injectables? *Only if 3b = Injectable ticked*	□□□

Completed by (initials) □□□□	**Date** □□/□□□/□□□□

FGE Follow-up genital examination

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Completed by the study clinician at follow-up visits. In the Reduced arm, only the external examination is done.

Participant ID	Site	Visit	Date
□ - □□□	□ 1 Dar □ 2 Mwanza □ 3 Mbeya	_____	□□/□□□/□□□□

1a	First day of last menstrual period	□□ / □□□ / □□□□ dd mmm yyyy
1b	Was a genital examination done?	□ Yes □ No → *If No, give the reason and go to item 6.*
1c	Reason the examination was not done *Only if 1b is No*	_____ _____

External examination

2 Were any of these abnormalities seen on external genital examination?

Abnormality on external examination	Answer
a. Abnormal vaginal discharge	□ Yes □ No □ Not examined
b. Genital warts	□ Yes □ No □ Not examined
c. Other	□ Yes □ No □ Not examined

Only if 1b is Yes

2c	Other abnormality, specify *Only if 2c is Yes*	_____
3	Does the participant have genital lesions consistent with herpes? *Only if 1b is Yes*	□ Yes □ No □ Not examined

Collect an HSV culture swab (item 5f).

Speculum examination (not in the Reduced arm)

4a	Was a speculum examination done?	□ Yes □ No → *Not in the Reduced arm. If No, go to item 5.*
4b	Cervical mucopus *Only if 4a is Yes*	□ Yes □ No □ Not examined
4c	Other abnormality on speculum examination *Only if 4a is Yes*	□ Yes □ No □ Not examined
4c	Other abnormality, specify *Only if 4c is Yes*	_____

Samples collected (not in the Reduced arm)

5a	HSV PCR swab collected?	□ Yes □ No □ Not applicable → *Not in the Reduced arm.*
5b	Vaginal swab for BV Gram stain collected? *Only if 1b is Yes*	□ Yes □ No □ Not applicable
5c	Vaginal swab for BV storage collected? *Only if 1b is Yes*	□ Yes □ No □ Not applicable
5d	Vaginal swab for GC, CT and TV collected? *Only if 1b is Yes*	□ Yes □ No □ Not applicable
5e	Cervicovaginal lavage collected? *Only if 1b is Yes*	□ Yes □ No □ Not applicable
5f	HSV culture swab collected? *Only if 1b is Yes*	□ Yes □ No □ Not applicable
5g	Was a blood sample for HIV and HSV ELISA collected?	□ Yes □ No

Laboratory results (aetiological diagnosis)

6 Laboratory tests done for the aetiological diagnosis of STIs

Test	Done?	Result
a. Gonorrhoea	□ Yes □ No	□ Negative □ Positive
b. Chlamydia	□ Yes □ No	□ Negative □ Positive
c. Trichomoniasis	□ Yes □ No	□ Negative □ Positive
d. Bacterial vaginosis	□ Yes □ No	□ Negative □ Positive

HSV-2 seroconversion		
7a	Since the last visit, has the participant seroconverted to HSV-2?	□ Yes □ No *→ If No, end of form.*
7b	Date of seroconversion *Only if 7a is Yes*	□□ / □□□ / □□□□ dd mmm yyyy

Completed by (initials) □□□□	**Date** □□/□□□/□□□□

MV Missed visit

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Completed by the study nurse when the window for a scheduled visit has closed. One form per missed visit.

Participant ID	Site	Visit missed	Date
□ - □□□	□ 1 Dar □ 2 Mwanza □ 3 Mbeya	Month ____	□□/□□□/□□□□

1	Why was the visit missed?	□ School commitments □ Work □ Parent or guardian refused □ Travelling or out of the area □ Ill or in hospital □ Transport or fare problems □ Refused the visit □ Moved away □ Could not be contacted □ In prison □ Other
1a	Other reason, specify *Only if 1 is Other*	_____
2	Comments	_____ _____

Completed by (initials) □□□□	**Date** □□/□□□/□□□□